



YACHAY TECH UNIVERSITY
 COURSE: OPTIMIZATION METHODS
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 PROJECT NO. 1
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In this project we put into practice the concepts learned about unconstrained optimization. Two sample exercises are presented to apply techniques and algorithms that we have developed in the course. For all effects the parameter $d\%$ is the last digit of your ID plus 2. if your ID is 1758245290, then $d = 2$.

A report is expected, with the following details:

- Introduction: General description of unconstrained optimization. Methods used for solving these problems. Objectives of this work.
- Methods. A complete description of the following methods: Steepest Descent, Conjugate Gradient, and Newton Method. Algorithms, and general issues about its convergence is expected.
- Experimental environment: Describes the variables needed for each method and its values, or initial values. Also, the test functions and parameters needed.
- Experimental results: A table for each method containing its first 10 iterations.
- Conclusions: Based on the experimental results generate a short discussion for each method based on the comparison of expected exact solution and the experimental results. Conclude with a brief recommendation according with these results.

I. Given the function

$$f(x_1, x_2) = (x_1 - d)^4 + (x_1 - 2dx)^2$$

- a) Program the Steepest Descent, Newton Method, and Conjugate Gradient Method for solving the problem: $\min f(x)$. Internal documentation is mandatory.
- b) Take $x_0 = [1; d]$ and apply Steepest descent, Newton Method, and Conjugate Gradient in order to find the minimum.
- c) Present a table with the gradients(or its norms) for the first 10 iterations.
- d) Present a graph with the relation between iterations versus error.

II. For the functions given, determine explicitly all the stationary points and use sufficiency conditions to check what kind of points they are.

1) $f = 3x_1 + \frac{100}{x_1x_2} + dx_2$

2) $f = (x_1 - d)^2 + x_1x_2 + (x_2 - d)^2$

Use one of the previous code to corroborate your findings.

ZC